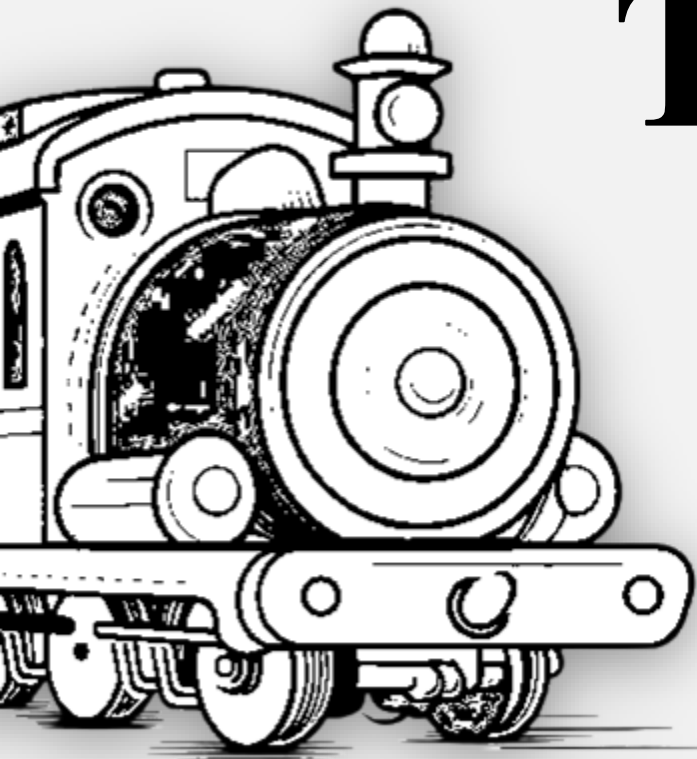




Ticket To Tech

*Group Members: Carrie Houston, Makenzie
Kadjeski, & Jeny Thomas*



Problem Overview

➤ Problem:

- Finding Optimal Path to Classes

➤ Algorithms Implemented in Python & C++:

- Breadth First Search (BFS)
- Depth First Search (DFS)
- A* Search
- Bidirectional Search

➤ The goal is to find the most efficient algorithm to solve this problem.

Objective & Research:

- Objective: Determine which algorithm is the most efficient for finding the optimal path between two places.
- Time complexity :
 - BFS, DFS, and A*: $O(|n|^2)$
 - Bidirectional: $O(|b|^{d/2})$
- Expected Results: Bidirectional will be the most efficient algorithm

Description of Experiment:

- Real World Datasets of Tennessee Tech's Campus
 - Small, medium, & large unweighted graphs
- Tested code on Makenzie's MacBook Pro five times for:
 - Each algorithm on each graph
 - Brute force on small and medium graphs
- Routes we tested
- Comparisons

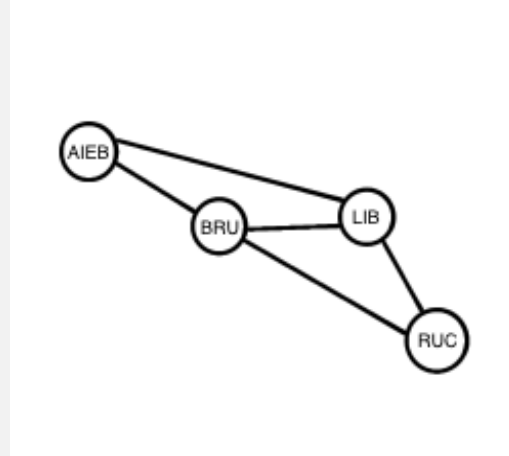
Input Matrices and Graphs

[0, 1, 1, 0]

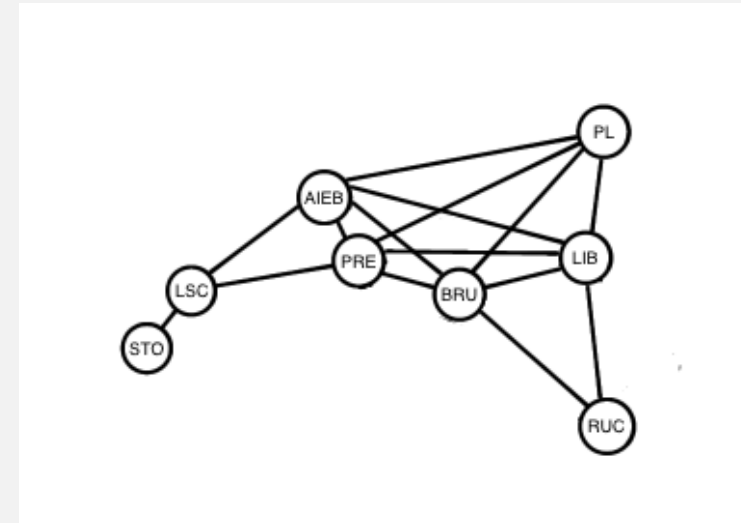
[1, 0, 1, 1]

[1, 1, 0, 1]

[0, 1, 1, 0]

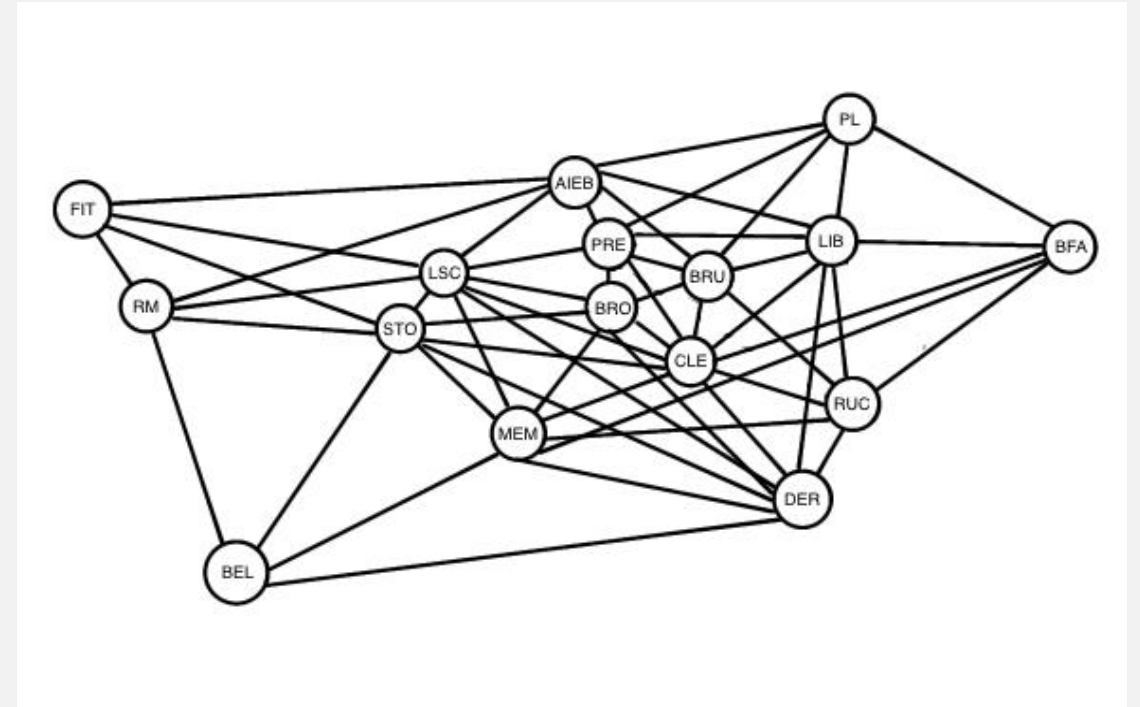


[0, 1,	1,	0,	1,	1,	1,	0],
[1, 0,	1,	1,	1,	1,	0,	0],
[1, 1,	0,	1,	1,	1,	0,	0],
[0, 1,	1,	0,	0,	0,	0,	0],
[1, 1,	1,	0,	0,	1,	0,	0],
[1, 1,	1,	0,	1,	0,	1,	0],
[1, 0,	0,	0,	0,	1,	0,	1],
[0, 0,	0,	0,	0,	0,	1,	0]



Input Matrices and Graphs

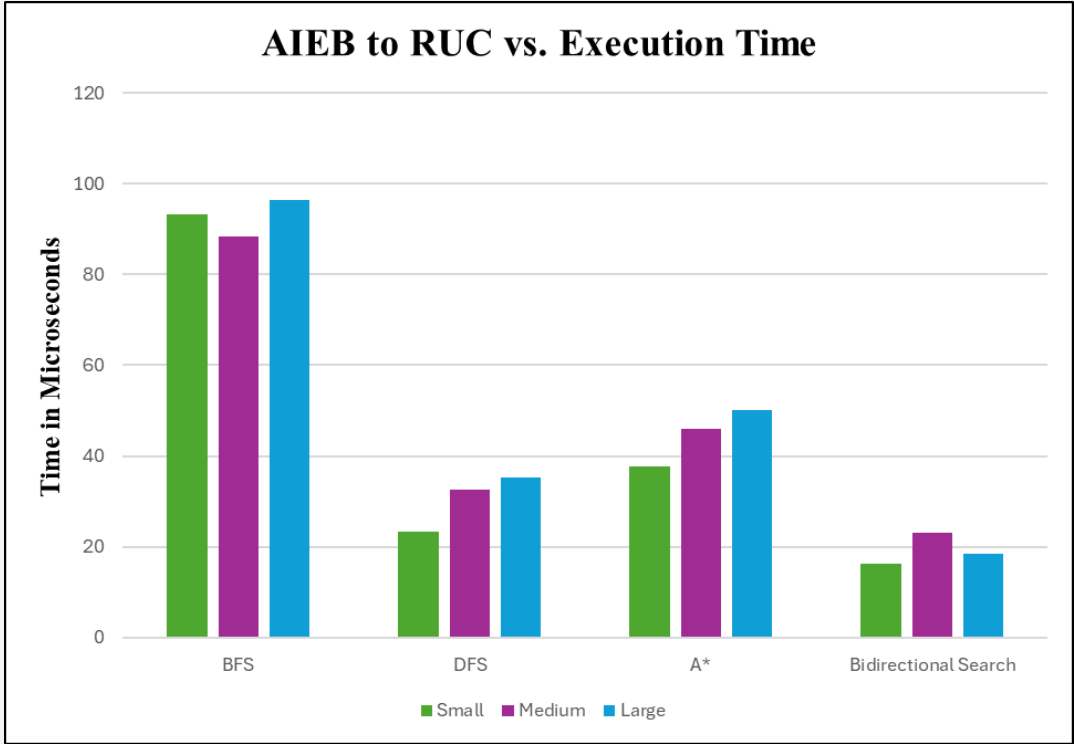
```
[0, 1, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1],  
[1, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0],  
[1, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0],  
[0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0],  
[1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0],  
[1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0],  
[1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1],  
[0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 1],  
[0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0],  
[0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0],  
[0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0],  
[0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0],  
[0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1],  
[0, 0, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0],  
[1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1],  
[1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0]
```



Results:

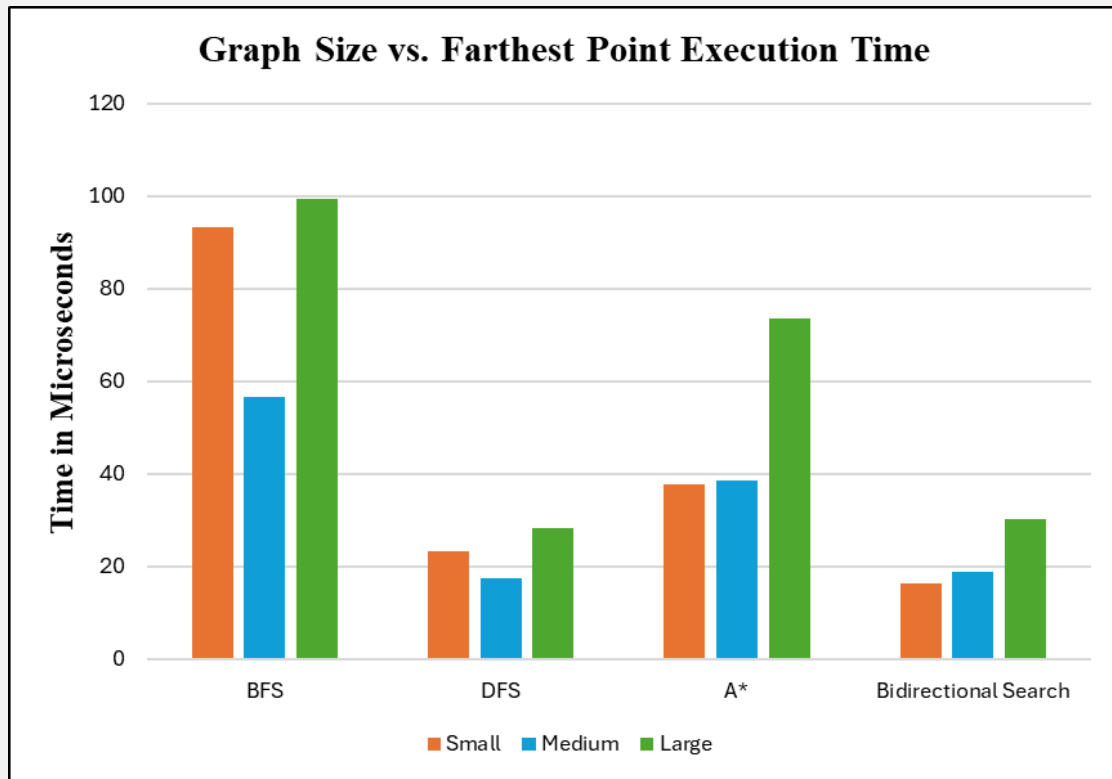
- Execution times ranked from lowest to highest:
 - Bidirectional
 - DFS
 - A*
 - BFS
- Most consistent algorithms:
 - BFS
 - Bidirectional

Figure 1 and Table 1



AIEB to RUC vs. Execution Time (μs)			
Algorithm	Small	Medium	Large
BFS	93.2	88.4	96.4
DFS	23.4	32.6	35.4
A*	37.8	46	50.2
Bidirectional Search	16.4	23.2	18.4

Figure 2 & Table 2

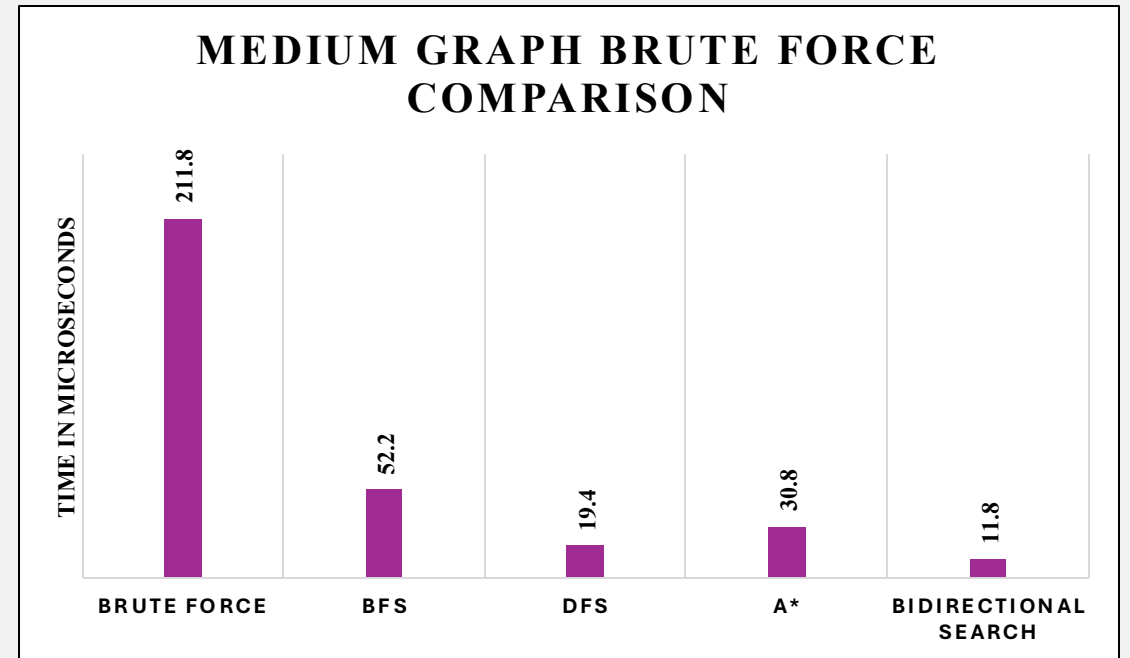
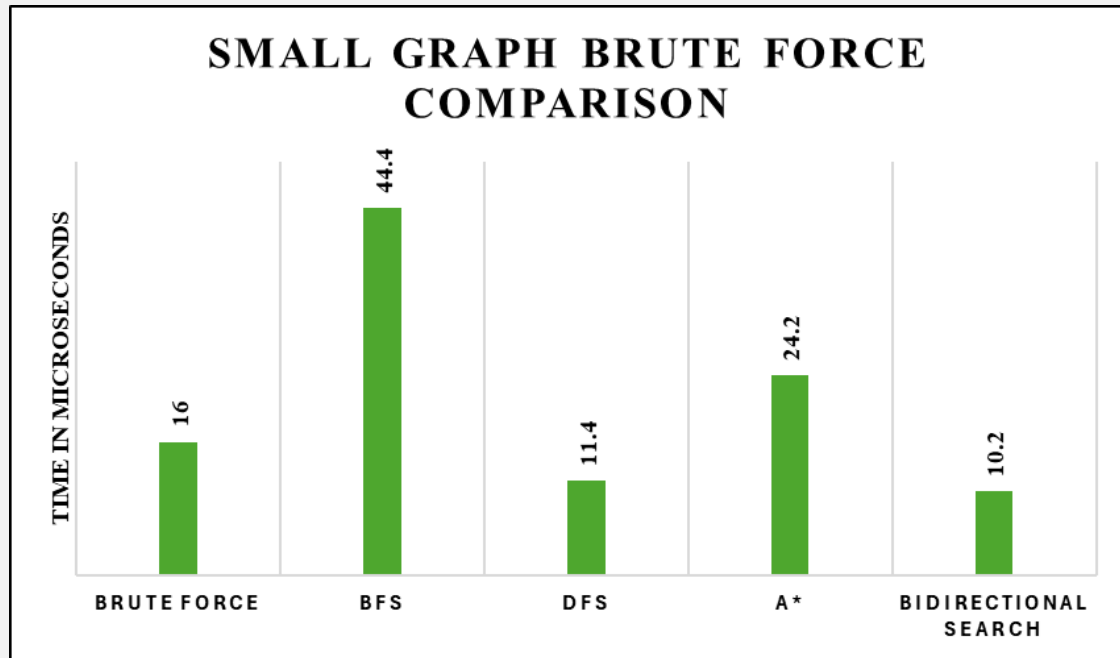


Graph Size vs. Farthest Point Execution Time (μs)			
Algorithm	Small	Medium	Large
BFS	93.2	56.6	99.4
DFS	23.4	17.6	28.2
A*	37.8	38.6	73.6
Bidirectional Search	16.4	18.8	30.2

Table 3:

Average of Brute Force Method vs. Path Planning Algorithms		
Algorithm	Small	Medium
Brute Force	16 μ s	211.8 μ s
BFS	44.4 μ s	52.2 μ s
DFS	11.4 μ s	19.4 μ s
A*	24.2 μ s	30.8 μ s
Bidirectional Search	10.2 μ s	11.8 μ s

Figures 3 & 4



Interpretation Summary:

- Ran the program 5 times per input
- Larger number of vertices → longer execution times
- Bidirectional was most efficient when searching for the path between the furthest points; BFS was least efficient
- Brute force is most efficient with small input sizes

Limitations & Next Steps:

- Limited input size, graph structure, limited time
- Larger input size, sparser graph, implement a weighted graph, measure different outputs

QR Code:



References:

- GeeksforGeeks. (2008). GeeksforGeeks.
<https://www.geeksforgeeks.org/> -used in objective/research
- Kumar, Rajesh. (2024). DataCamp. The A* Algorithm: A Complete Guide | DataCamp
- Levitin, A. (2012). *Introduction to the design & analysis of algorithms*. Pearson. -used in objective/research
- OpenAI. (2025). *ChatGPT* [Large language model]. OpenAI.
<https://chat.openai.com> -experimentation
- Radian, C. (2025). *Section 3.5 DFS_BFS Presentation*. -used in objective/research